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Age-Related Scattered Hypofluorescent Spots as an Adverse Prognostic Factor for Polypoidal Choroidal Vasculopathy

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Purpose: Polypoidal choroidal vasculopathy (PCV) demonstrates significant prognostic variability, and the impact of age-related scattered hypofluorescent spots observed in late-phase indocyanine green angiography (ASHS-LIA) on the prognosis of PCV remains under-researched. This study aims to investigate the association between ASHS-LIA in PCV and prognosis using the AdaBoost machine learning model.

Design: A cross-sectional study.

Participants: The study included patients diagnosed with PCV and treated with anti-VEGF therapy at 2 medical institutions between 2012 and 2021.

Methods: We conducted a retrospective analysis of the clinical characteristics, anti-VEGF treatment history, and outcomes of the participants, classifying them based on the presence or absence of ASHS-LIA. An AdaBoost meta-estimator was applied to predict prognosis, including disease stability, injection frequency, and time to first remission, utilizing features selected through principal component analysis.

Main Outcome Measures: The prognostic significance of ASHS-LIA was assessed by feature importance, with the mean decrease in impurity serving as the evaluation metric.

Results: Of 57 eyes with PCV, 31 exhibited ASHS-LIA and 26 did not. Compared with the non-ASHS-LIA group, the ASHS-LIA group had fewer patients who achieved a super-stable status without recurrence for >18 months post-remission ($P = 0.03$), required a longer time to reach first remission ($P = 0.04$), and needed more injections ($P < 0.001$). AdaBoost models confirmed the importance of ASHS-LIA for predicting disease stability, injection demand, and time to first remission, ranking it as the third, seventh, and eighth top contributory factor, respectively.

Conclusions: Machine learning analysis identified ASHS-LIA as a negative prognostic factor in PCV, correlating with reduced disease stability, higher recurrence rates, and increased treatment requirements. These findings suggest that ASHS-LIA could serve as a valuable marker for assessing prognosis and guiding treatment strategies in PCV management.

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Polypoidal choroidal vasculopathy (PCV) is a distinct subtype of neovascular age-related macular degeneration (AMD), which predominantly affects individuals of Asian descent.[1,2] Polypoidal choroidal vasculopathy is characterized by the presence of polypoidal lesions and abnormal branching vascular networks within the choroid, leading to recurrent episodes of subretinal hemorrhage, exudation, and vision loss. Early detection and an accurate diagnosis of PCV are crucial for initiating timely interventions and optimizing patient outcomes.

Polypoidal choroidal vasculopathy exhibits a relatively significant variation in prognosis compared with other subtypes of AMD,[3–5] which highlights the need to identify imaging biomarkers capable of predicting PCV progression in the context of treatment planning. Indocyanine green angiography (ICGA), a commonly employed imaging technique, has revealed the presence of hypofluorescent spots in its late phase, indicating lipid accumulation in Bruch membrane.[6,7] These hypofluorescent spots on late-phase ICGA images have been termed "age-related scattered hypofluorescent spots on late-phase ICGA" (ASHS-LIA). Notably, ASHS-LIA in PCV eyes has been associated with bilateral involvement, reduced choroidal vasculature hyperpermeability, and a thicker choroid. Nevertheless, the relationship between ASHS-LIA and the prognosis of eyes with PCV remains largely unexplored.

In this study, we aimed to explore the association between ASHS-LIA observed in PCV eyes and the prognostic impact of this condition. To achieve this objective, an AdaBoost classifier/regressor, a widely used meta-estimator

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in machine learning (ML), was employed.[8,9] This technique was utilized to construct an ML model that uses a limited dataset and can incorporate diverse clinical features, including the presence of ASHS-LIA. By utilizing these features to create a model that can accurately predict the prognosis of PCV, we aimed to assess the ability of ASHS-LIA to influence the model's performance. Ultimately, our objective was to determine the utility of ASHS-LIA for predicting PCV prognosis using ML techniques. If ASHS-LIA is identified to be a crucial predictor, its presence is expected to play a pivotal role in the establishment of individualized treatment plans.

Methods

Figure 1 presents a flowchart outlining the selection process for the study population and the ML analysis process, including principal component analysis (PCA) and the AdaBoost meta-estimator.

Study Population and Data Source

We conducted a retrospective analysis of the medical records of patients diagnosed with PCV who underwent intravitreal injections at the ophthalmology departments of Severance Hospital and National Health Insurance Ilsan Hospital between January 1, 2012, and December 31, 2021.

The diagnosis of PCV was established by 2 independent ophthalmologists (E.Y. Choi and S.H. Kim) based on criteria provided by the CONAN group.[10] According to these criteria, PCV is characterized by the presence of a polypoidal lesion with a branching vascular network observed on ICGA accompanied by exudation or hemorrhages. The inclusion criteria for this study included patients with PCV aged 50 years or older who had not received any prior treatment, who underwent injection therapy following the pro re nata protocol, and who had a minimum follow-up period of 18 months after receiving 3 initial loading injections. Eyes with concurrent conditions such as high myopia (spherical equivalent less than −6.0 diopters or an axial length greater than 26.0 mm), glaucoma, uveitis, diabetic retinopathy, retinal vascular occlusion, recent eye trauma, or intraocular surgeries during the follow-up period (including cataract surgery), and insufficient late-phase ICGA results were excluded from the analysis.

Before initiating anti-VEGF treatment, all patients underwent a comprehensive evaluation including a detailed medical and ophthalmic history, best-corrected visual acuity testing, intraocular pressure assessment, fundoscopic examination through indirect ophthalmoscopy, and an anterior segment examination via slit-lamp biomicroscopy. Spectral-domain OCT in the enhanced depth imaging mode was performed using a Spectralis system (Heidelberg Engineering). Fluorescein angiography and ICGA were performed using a Heidelberg Spectralis HRA (Heidelberg Engineering). For angiography, a 30° field fundus camera was used to capture the macula and optic disc, focusing on the posterior pole. To obtain images of the mid-periphery, multiple images were taken with off-center fixation, ensuring overlapping fields.

The institution that provided data for this study followed the unique Korean national insurance system, which imposes certain restrictions on the use of anti-VEGF drugs in the treatment of AMD. Therefore, the choice of treatment agents varied according to insurance criteria, and the study included cases of bevacizumab alone, ranibizumab alone or in combination with bevacizumab, or aflibercept alone or in combination with bevacizumab. There were no cases of ranibizumab and aflibercept combined due to insurance coverage restrictions. All patients received a standard initial loading dose of intravitreal anti-VEGF, administered 3 times at 1-month intervals, followed by treatments until they achieved remission, defined as a dry macula on OCT with no subretinal fluid, intraretinal fluid, subretinal or intraretinal hemorrhage, or subretinal turbid exudation, indicating active neovascularization.[11] Injections were performed even if only one of these findings was present, regardless of the fluid amount, even if minimal. The change in pigment epithelial detachment (PED) and subretinal hyperreflective material was not considered an indication for injection. If there was no disease activity on OCT,

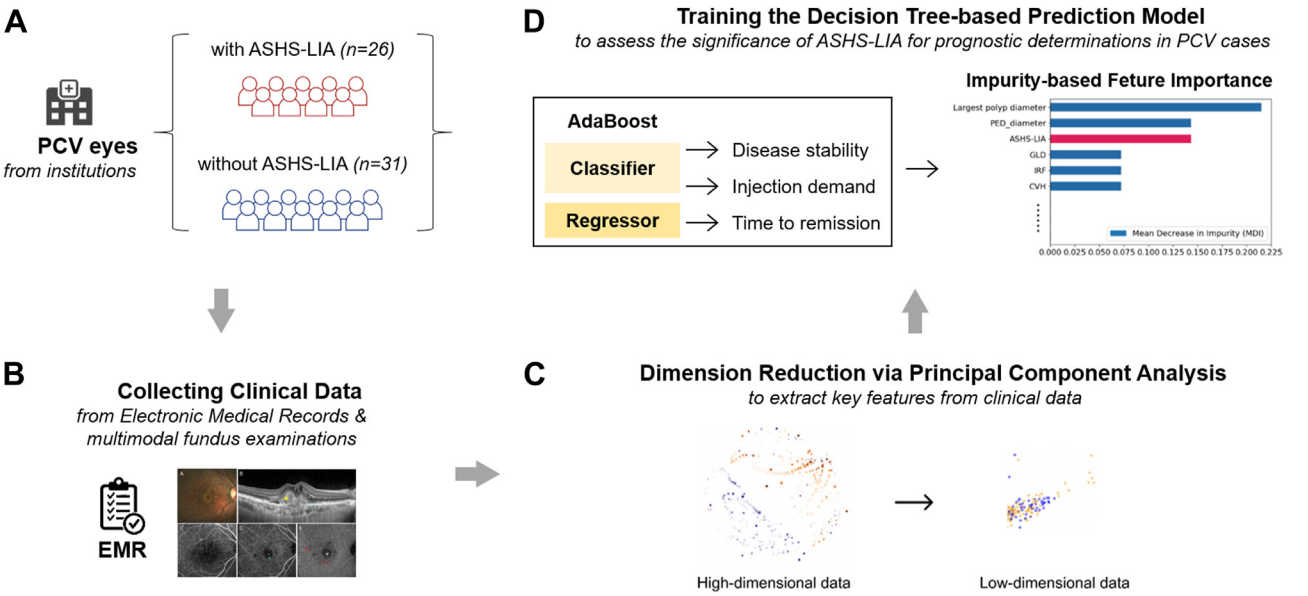


Figure 1. Flow diagram (**A**–**D**) of the retrospective study process. ASHS-LIA = age-related scattered hypofluorescent spots on late-phase indocyanine green angiography; EMR = electronic medical records; PCV = polypoidal choroidal vasculopathy.

injections were not administered, but regular follow-ups were typically continued at 4-week to 6-week intervals. Visual acuity, intraocular pressure, fundus photography, and spectral-domain OCT examinations were conducted at each visit.

Age-related scattered hypofluorescent spots observed in late-phase indocyanine green angiography was defined as the presence of hypofluorescent spots observed on late-phase ICGA, with no corresponding abnormalities on other multimodal imaging forms, including fundus photography, autofluorescence, fluorescein angiography, or spectral-domain OCT (Figs 2 and 3).[6,7] Because the ICGA delay phase is typically defined as 15 minutes after dye injection, hypofluorescent spots observed starting 15 minutes postinjection were considered ASHS-LIA.

The patients were divided into 2 groups based on the presence or absence of ASHS-LIA. We compared the basic characteristics and anatomical differences between these groups after the 3 initial loading treatments. Foveal configurations were confirmed using OCT. Central retinal thickness and subfoveal choroidal thickness were measured using a horizontal line scan through the center of the fovea. Subretinal fluid, intraretinal fluid, and the diameter and height of PED were identified. The number and characteristics of polypoidal lesions and the greatest linear dimension of the PCV complex were assessed using early- to mid-phase ICGA and OCT images, while lipid exudation and subretinal hemorrhages were detected using OCT and color fundus photography. Furthermore, we analyzed the number of injections required for first remission, the total number of injections administered during the follow-up period, and the period from the first remission to recurrence. First remission was defined as the first occurrence of dry macula on OCT after initiating anti-VEGF therapy. Remission was defined as the absence of fluid, hemorrhage, or exudation, indicating a lack of disease activity. A "super-stable status" was defined as maintaining remission without recurrence for >18 months, and patients who achieved this status were classified into the super-stable PCV group.[3] Additionally, we defined "injection demand" as the average annual demand for anti-VEGF injections and divided the patients into 2 groups, as follows: those who required >7 injections per year and those who required ≤7 injections per year. We categorized patients based on the time to first remission into 2 groups, according to whether it exceeded 7 months or not.

Ophthalmic imaging findings were analyzed independently by 2 retinal specialists (with 7 and 8 years of experience, respectively) who were blinded to the clinical data. Therefore, all measurements were performed twice to evaluate intergrader agreement. For continuous measurements, the average value was used in the analysis. For categorical measurements, cases where the results did not match between graders were deemed inconclusive and excluded from the analysis.

This study adhered to the principles outlined in the Declaration of Helsinki and was conducted in accordance with the guidelines of our institutional review board (No. 2022-11-046). The requirement for informed consent was waived owing to the retrospective nature of the study.

Statistical Analyses

Statistical analyses were performed using commercial software (SPSS, version 22.0; SPSS Inc). Best-corrected visual acuity was measured using a Snellen chart and converted to the logarithm of the minimal angle of resolution for statistical analyses. Quantitative values are presented as the mean ± standard deviation. To compare quantitative data, such as age, best-corrected visual acuity, central

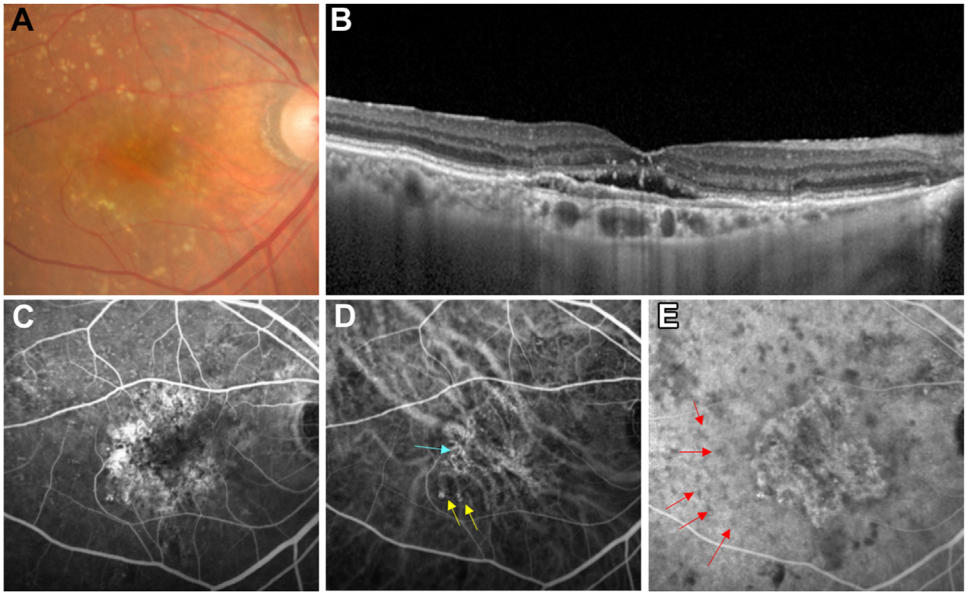


Figure 2. Multimodal imaging of a PCV patient with age-related scattered hypofluorescent spots on late-phase indocyanine green angiography. Initial findings from the first visit of an 82-year-old female patient with PCV. **A,** Extramacular drusen is observed in the fundus photograph of the right eye, accompanied by subretinal exudation and yellowish subretinal lesions around the macula. **B,** OCT reveals subretinal fluid and fibrovascular pigment epithelial detachment in the macular region, along with intraretinal hyperreflective material. **C,** Initial fluorescein angiography shows macular leakage and staining. **D,** Initial indocyanine green angiography substantiates the diagnosis of PCV by illustrating a polyp (yellow arrows) and a branching vascular network (blue arrow) in the macular region. **E,** Late-phase indocyanine green angiography showing multiple hypofluorescent spots (red arrows) encircling the PCV lesion in the macular region, suggestive of age-related scattered hypofluorescent spots on late-phase indocyanine green angiography. PCV = polypoidal choroidal vasculopathy.

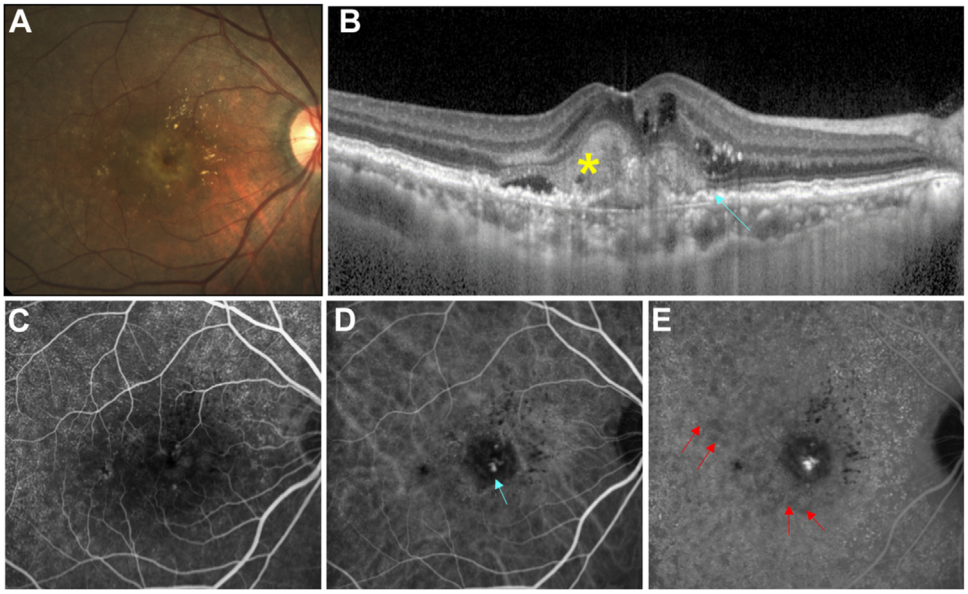


Figure 3. Multimodal imaging of a PCV patient with age-related scattered hypofluorescent spots on late-phase indocyanine green angiography. Examination results from the initial visit of a 78-year-old male patient with PCV. **A,** Fundoscopic examination reveals turbid and hard exudates in the macular region. **B,** OCT reveals turbid subretinal fluid (yellow asterisk) and intraretinal fluid along with an irregular fibrovascular pigment epithelial detachment (blue arrow) in the macular region. **C,** Initial fluorescein angiography shows uncertain leakage and staining of the macular region. **D,** Initial indocyanine green angiography confirms the diagnosis of PCV by revealing a polypoidal lesion with hyperfluorescence and a halo in the surrounding area (blue arrow). **E,** Late-phase indocyanine green angiography shows hypofluorescent spots surrounding the PCV lesion (red arrow), suggestive of age-related scattered hypofluorescent spots on late-phase indocyanine green angiography. PCV = polypoidal choroidal vasculopathy.

retinal thickness, subfoveal choroidal thickness, PED diameter and height, the greatest linear dimension of the PCV complex, and the length of the nonrecurrence period, we employed the independent 2-sample t test and Mann–Whitney U test. Chi-square and Fisher exact tests were used to compare categorical variables. Multivariate logistic regression analysis was conducted for each of the 3 dependent variables (disease stability, injection demand, and time to first remission), incorporating factors identified as significant ($P < 0.15$) in the univariate analysis. Statistical significance was set at a P value of <0.05.

ML-Based Analyses

Machine learning has recently gained considerable attention in the medical domain because ML-based regressors and classifiers are commonly used to support clinical findings or address result analyses. In this part of our study, we trained ML models for the following prediction tasks to analyze the impact of ASHS-LIA on the model's predictions in comparison to other clinical features. During model training, overfitting was intentionally permitted to achieve near-perfect predictors.

- Disease stability (task 1): a binary classification task in which the model predicts whether a patient belongs in the super-stable PCV group.
- Injection demand (task 2): a binary classification task in which the model predicts whether a patient requires at least 7 anti-VEGF injections per year.
- Time to remission (task 3): a regression task in which the model predicts the time (months) to first remission of a given patient.

In this section, we describe how we preprocessed clinical data, trained the ML models, and measured the feature importance of ASHS-LIA.

Dimensional Reduction of Clinical Features

It is suboptimal and nontrivial to develop ML models with a dataset that is small in size yet high in dimension[12] (i.e., having a large number of features [our clinical data had 28 types of features in total, including ASHS-LIA]). Therefore, we first selected baseline demographic information and all relevant clinical data related to the prognosis of PCV and then applied PCA to transform the original high-dimensional data into low-dimensional data with features that better captured the variance in the original data.[13] Values of features were standardized within each feature before PCA. The implementation details of the manual feature removal and PCA for each task are provided in Methods S1 (available at www.ophthalmologyscience.org).

ML Models to Measure the Feature Importance of ASHS-LIA

Using the processed data, we trained the AdaBoost models on the aforementioned tasks and measured the contribution of ASHS-LIA to model predictions. AdaBoost[14] is a decision tree–based ML algorithm that has been widely applied to diverse research areas, including ophthalmology.[8,9] One of its uses is to assist in selecting important features from clinical data[15–18] through impurity-based feature importance (i.e., the mean decrease in impurity, known as the Gini importance score).[19] The mean decrease in impurity is associated with how often a feature is used as a criterion to split the decision nodes in a tree and how it reduces the impurity of nodes. Formally, we first used the processed data to train the AdaBoost classifier (for tasks 1 and 2) and then the AdaBoost regressor (for task 3) to overfit the prediction tasks (i.e., to achieve a near-perfect prediction performance based on the R^2 score [i.e., the coefficient of determination]). Subsequently, we computed and ranked the mean decrease in impurity of each feature

to assess the effect of ASHS-LIA on model predictions. Details of the AdaBoost models and mean decrease in impurity are provided in Methods S2 (available at www.ophthalmologyscience.org).

Results

We reviewed a total of 122 eyes diagnosed with PCV during the study period. We excluded 9 eyes previously treated elsewhere and 5 eyes that did not meet the 18-month follow-up requirement after the first remission. Additionally, 47 eyes with insufficient late-phase ICGA results were excluded. After assessing the presence of hypofluorescent spots on the latest available ICGA images (between 15 and 20 minutes) for the remaining 61 eyes, 4 cases with discrepancies in assessment between graders were deemed ungradable and excluded. Therefore, this study analyzed 57 eyes from 57 patients diagnosed with PCV. Of the 57 eyes, 26 had no hypofluorescent spots in the late phase of ICGA (non-ASHS-LIA group), and 31 had hypofluorescent spots (ASHS-LIA group). Table 1 presents the clinical characteristics of the ASHS-LIA and non-ASHS-LIA groups before treatment initiation. Although the average age and proportion of male patients in the ASHS-LIA group were slightly higher than those in the non-ASHS-LIA group, these differences were not statistically significant. Both groups had an average follow-up period of >3 years, and there was no difference in the follow-up duration between the 2 groups. Significantly more patients in the non-ASHS-LIA group achieved a super-stable status compared with those in the ASHS-LIA group ($P = 0.03$). There were no significant differences in the proportion of patients achieving first remission, the proportion of patients experiencing recurrence, or the OCT results between the 2 groups. However, a trend toward a thinner central retinal thickness and a higher frequency of exudation were observed in the ASHS-LIA group ($P = 0.09$ and $P = 0.09$, respectively).

The changes in clinical features after anti-VEGF treatment of the 2 groups were analyzed in detail and are listed in Table 2. One month after the 3 initial loading injections, there were no significant differences in the degree of improvement of the best-corrected visual acuity or the variables identified on OCT between the 2 groups. However, in comparison to the non-ASHS-LIA group, the ASHS-LIA group took longer to achieve first remission (5.54 months vs. 10.94 months, $P = 0.04$) and required more injections during the entire follow-up period (9.69 injections vs. 19.36 injections, $P < 0.01$). Additionally, although not statistically significant, there was a tendency for the period from first remission to recurrence to be relatively short ($P = 0.09$). Throughout the follow-up period, all patients received ranibizumab, aflibercept, or bevacizumab, either individually or in combination, in accordance with South Korea's specialized medical insurance system and insurance coverage standards. There were no significant differences in the drugs administered between the 2 groups.

In the binary logistic analysis of disease stability, ASHS-LIA's impact on disease stability was not statistically significant ($P = 0.24$), whereas female sex was significantly associated with a higher disease stability ($P = 0.043$, odds ratio: 3.639). Age-related scattered hypofluorescent spots observed in late-phase indocyanine green angiography's influence on injection demand (over 7 times per year) was suggested by a P value of 0.08. Sex also significantly affected injection requirements ($P = 0.04$, odds ratio: 0.287), with women requiring fewer injections. Age-related scattered hypofluorescent spots observed in late-phase

Table 1. Comparison of Baseline Clinical Characteristics of ASHS-LIA vs. Non-ASHS-LIA Patients with PCV

	ASHS-LIA (n = 31)	Non-ASHS-LIA (n = 26)	P Value
Age (yrs)	70.39 ± 7.46	66.65 ± 9.79	0.11
Sex (male/female)	23/8	14/12	0.09
Laterality (right/left)	19/12	15/11	0.50
Baseline BCVA (logMAR [decimal equivalent])	0.62 ± 0.49 (0.24 ± 0.32)	0.56 ± 0.45 (0.28 ± 0.35)	0.67
Total follow-up period (mos)	54.2 ± 27.8	44.3 ± 20.8	0.14
Greatest linear dimension (μm)	3496.8 ± 1730.9	3036.0 ± 177.6	0.33
Central retinal thickness (μm)	267.3 ± 141.2	329.73 ± 128.1	0.09
Subfoveal choroidal thickness (μm)	308.6 ± 107.2	324.8 ± 97.7	0.56
PED height (μm)	282.2 ± 244.8	375.7 ± 241.7	0.15
PED diameter (μm)	1390.4 ± 1137.0	1355.3 ± 776.9	0.89
Polyp number (single/multiple)	20/11	12/14	0.29
Intraretinal fluid*	7 (22.6)	2 (7.7)	0.12
Subretinal fluid*	27 (87.1)	23 (88.5)	0.60
Subretinal hemorrhage*	11 (35.5)	10 (38.5)	0.47
Lipid exudation*	9 (29.0)	3 (11.5)	0.09
Soft/hard drusen*	4/2 (19.4)	2/0 (7.7)	0.27
Choroidal vascular hyperpermeability	17 (54.8)	10 (38.5)	0.17

ASHS-LIA = age-related scattered hypofluorescent spots on late-phase indocyanine green angiography; BCVA = best-corrected visual acuity; logMAR = logarithm of the minimum angle of resolution; PCV = polypoidal choroidal vasculopathy; PED = pigment epithelium detachment.
Quantitative data are expressed as mean ± standard deviation.
*Categorical variables are expressed as numbers of existence with percentages in parentheses.

Table 2. Comparison of Clinical Characteristics of ASHS-LIA vs. Non-ASHS-LIA Patients with PCV after Anti-VEGF Therapy

	ASHS-LIA (n = 31)	Non-ASHS-LIA (n = 26)	*P* Value
After 3 initial anti-VEGF loadings			
BCVA change (logMAR [decimal equivalent])	−0.07 ± 0.30 (−0.85 ± 0.50)	−0.15 ± 0.40 (−0.71 ± 0.40)	0.43
Intraretinal fluid*	3 (9.7)	1 (3.9)	0.62
Subretinal fluid*	13 (41.9)	6 (23.1)	0.16
Subretinal hemorrhage*	5 (16.1)	3 (11.5)	0.72
Lipid exudation*	8 (25.8)	5 (19.2)	0.75
Central retinal thickness (μm)	170.7 ± 64.5	181.7 ± 74.8	0.56
Subfoveal choroidal thickness (μm)	289.3 ± 97.2	286.7 ± 80.0	0.91
PED height (μm)	163.3 ± 111.5	204.4 ± 173.4	0.29
PED diameter (μm)	1198.0 ± 1151.6	1099.2 ± 744.8	0.70
Injections required for first remission	6.37 ± 6.9	4.27 ± 3.33	0.12
Time to achieve first remission (mos)	10.9 ± 18.8	5.54 ± 6.81	**0.04**
Time to recurrence from first remission (mos)	15.0 ± 18.8	24.4 ± 21.9	0.09
Super-stable PCV*	8 (25.8)	14 (53.8)	**0.03**
Total injection numbers	19.4 ± 13.9	9.7 ± 9.5	**<0.001**
Types of anti-VEGF agents*			0.21
Bevacizumab alone	5 (16.7)	8 (30.8)	
Ranibizumab ± Bevacizumab	14 (46.7)	10 (38.5)	
Aflibercept ± Bevacizumab	11 (36.7)	8 (30.8)	

ASHS-LIA = age-related scattered hypofluorescent spots on late-phase indocyanine green angiography; BCVA = best-corrected visual acuity; logMAR = logarithm of the minimum angle of resolution; PCV = polypoidal choroidal vasculopathy; PED = pigment epithelial detachment.
Quantitative data are expressed as mean ± standard deviation. Bold in the *P* value indicates statistical significance.
*Categorical variables are expressed as numbers of existence with percentages in parentheses.

indocyanine green angiography was not a significant factor in determining the time to first remission either. For the detailed binary logistic regression results, please refer to Table S3 (available at www.ophthalmologyscience.org).

We report the experimental results of our ML-based analyses regarding the feature importance of ASHS-LIA in Figure 4. The models demonstrated nearly perfect mapping of input features to the ground-truth labels of targets, with R^2 scores of 1.000, 1.000, and 0.942 for tasks 1, 2, and 3, respectively. For task 1, in which the model was trained to predict whether a patient belongs in the super-stable PCV group, ASHS-LIA ranked third in feature importance (i.e., contributing to model predictions) out of the 14 features used for prediction. For task 2, in which the target for prediction was the injection demand (i.e., >7 times or not) for each given patient, ASHS-LIA ranked seventh out of 17 features. Finally, ASHS-LIA ranked eighth out of 11 features for task 3, in which the model was trained to predict the time to first remission for a given patient. The feature importance of ASHS-LIA was confirmed through a twofold cross-validation with a smaller data size (number of data = 30), as shown in Figure S5 (available at www.ophthalmologyscience.org).

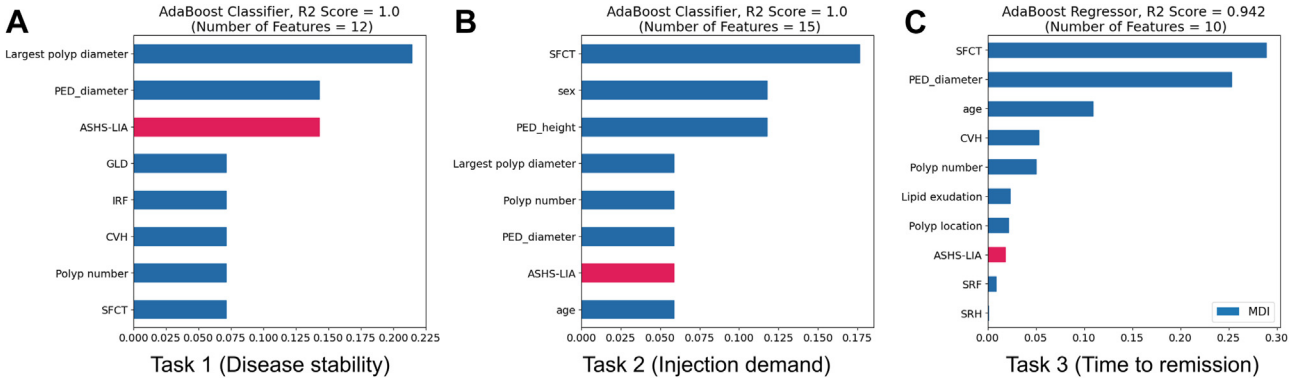


Figure 4. Feature importance of each variable ranked by the mean decrease in impurity AdaBoost classifier for (**A**) disease stability (Task 1) and (**B**) injection demand (Task 2); (**C**) AdaBoost regressor for time to remission (Task 3). ASHS-LIA, age-related scattered hypofluorescent spots on late-phase indocyanine green angiography; CVH = choroidal vascular hyperpermeability; GLD = greatest linear dimension; IRF = intraretinal fluid; PED = pigment epithelial detachment; SFCT = subfoveal choroidal thickness; SRF = subretinal fluid; SRH = subretinal hemorrhage.

Discussion

In this study, we employed ML techniques to elucidate the prognostic significance of ASHS-LIA in eyes with PCV. Age-related scattered hypofluorescent spots observed in late-phase indocyanine green angiography was associated with a lower stability (more frequent recurrence) and a higher demand for injections in eyes with PCV. This study contributes significantly to the literature in 2 ways. First, for the first time, we demonstrated that ASHS-LIA serves as a poor prognostic factor in this patient population. Consequently, we anticipate that this finding will be instrumental in tailoring more individualized treatment plans for patients with PCV, especially considering the variability in prognosis for this disorder compared with other types of AMD. Second, we used ML to identify a novel prognostic factor that evaded detection during logistic regression analysis, a more conventional statistical method. This process was achieved through a crucial feature-selection process utilizing an overfitting AdaBoost model. The construction of this model followed the simplification of the influence of various features through dimensional reduction using PCA.

Traditionally, PCV has been considered to be more resistant to anti-VEGF therapy than other types of neovascular AMD.[4,5] However, a subset of PCV, termed "super-stable PCV," has shown a notably favorable response to anti-VEGF treatment. Patients in this super-stable category are generally younger, predominantly female, and exhibit a singular, smaller polyp at baseline.[3] Interestingly, the presence of ASHS-LIA, a potential correlate of Bruch membrane lipid accumulation, is associated with the occurrence of soft drusen.[6] Further, ASHS-LIA is suggested to be a precursor lesion for PCV and has been linked to bilateral involvement of PCV, less choroidal vascular hyperpermeability, and a thicker choroid.[7]

To the best of our knowledge, this study represents the first investigation of the association between ASHS-LIA in eyes with PCV and clinical outcomes. Our findings reveal that the presence of ASHS-LIA is correlated with a lower stability of lesions and a higher injection demand, suggesting that the identification of ASHS-LIA may facilitate a more customized treatment strategy for patients with PCV. Further research in this area could refine the therapeutic approaches for PCV based on individual ASHS-LIA grades.

Numerous studies have employed ML technology to construct models for diagnosing ophthalmological diseases and predicting prognosis based on multimodal clinical data, presenting key features that are crucial for model performance.[20–22] However, the application of ML to identify novel features that are readily applicable to clinical practice is limited. One example is for colorectal cancer, in which the tumor adipose feature, a newly derived histopathological feature identified through ML, demonstrated prognostic value for the disease[23–25] and was considered usable by pathologists.[26] Previous studies have noted that ASHS-LIA is a characteristic early finding in some PCV patients,[7,27,28] although its impact on prognosis remains unclear. Using more conventional statistical analyses, we could not confirm a significant association between the presence of ASHS-LIA and disease prognosis. However, by using PCA to extract low-dimensional information from high-dimensional data and construct an overfitting AdaBoost model, we were able to successfully identify the prognostic value of ASHS-LIA. This approach may serve as a universal method for medical tasks, facilitating the discovery of novel diagnostic and prognostic factors within small datasets through the characterization of the complex influence of multiple factors.

In our study, those without ASHS-LIA demonstrated a significantly more stable course, with twice as many patients experiencing long-term remission as those in the ASHS-LIA group. Machine learning analysis identified ASHS-LIA to be the third most important factor affecting PCV stability, along with a large polyp size and PED size. Factors associated with super-stable PCV, including a younger age, single polyp, and small PED, have been reported,[3] and ASHS-LIA absence has emerged as a new identifying factor. Furthermore, ASHS-LIA was found to be the seventh and eighth most crucial factor for determining the need for injections and the remission time in our study. Additionally, the subfoveal choroidal thickness and PED size emerged as the most important factors influencing both injection demand and time to remission. Pigment epithelial detachment reflects the exudate in polypoidal lesions,[29] and choroidal thickness indicates the choroid capillary permeability,[30] with both influencing PCV recurrence and visual prognosis. Notably, ASHS-LIA was linked to an increased need for anti-VEGF administration (1.5 times more) and a longer time to remission (twice as long), suggesting its association with more frequent PCV lesion recurrence and a poor prognostic outcome. These findings highlight the importance of ASHS-LIA for understanding PCV stability, recurrence, and treatment response.

This study had some limitations. First, selection bias and clinical heterogeneity may have occurred in the retrospective analysis of a small number of subjects. In particular, the pattern of injection treatment and type of medication among the comparison groups could not be completely controlled. Additionally, as this study was based in a clinical setting, follow-up visits were not uniformly conducted at monthly intervals. Due to patient conditions and practical constraints, the monitoring interval was occasionally extended to 2 to 4 months. However, to minimize clinical heterogeneity between the comparison groups, this study included only patients treated based on the pro re nata protocol. This allowed us to compare PCV disease activity based on injection number and the proportion of super-stable PCVs as the overall follow-up period between the groups was not significantly different. Additionally, there was no statistically significant difference in the types of anti-VEGF drugs administered between the 2 groups, and a recent study found no significant differences between long-term treatment outcomes with ranibizumab and aflibercept.[31,32] Moreover,

ASHS may have been undervaluated in this study due to the assessment being restricted to the 15- to -20 minute window. While Chen et al suggested assessing ASHS 20 to 40 minutes post-injection,[6,7] the late phase of ICGA generally begins at 15 minutes and our imaging was performed up to 20 minutes due to institutional protocols. Future studies with controlled prospective designs and larger sample sizes are needed to accurately evaluate the role of ASHS-LIA as a prognostic factor. Second, although we assessed the impact of ASHS-LIA on model predictions within our dataset (including cross-validation), external validation on other datasets was not performed. We recommend conducting external validation to further confirm the association between ASHS-LIA and prognoses. However, considering our study's objective of identifying new risk factors rather than developing successful ML prediction models, a potentially overfitted model could still be valuable. Third, while ML can measure feature importance, the direction of each feature's impact on the target task is not directly ascertainable, and the estimation relies on a basic statistical comparison of the dataset. Lastly, there is currently no direct histopathologic evidence that ASHS-LIA represents lipids in Bruch membrane and no direct clinical evidence demonstrating a substantial impact of ASHS-LIA on the visual outcome of eyes with PCV. Despite these limitations, our study offers additional insights into the prognosis of patients with PCV, potentially informing the development of more individualized treatment plans.

In conclusion, by employing ML techniques, ASHS-LIA, which frequently occurs in patients with PCV, was shown to play an important role in PCV prognosis. Patients with PCV associated with ASHS-LIA exhibited higher injection requirements, a longer time to remission, and a lower disease stability. Utilizing dimensional reduction (PCA) and decision tree models (AdaBoost) for small clinical datasets involving multiple factors can aid in the identification of new risk factors.

Data Statement

Data presented in the article or its supplementary materials are accessible within the publication, while additional data are not available.

Footnotes and Disclosures

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HUMAN SUBJECTS: Human subjects were included in this study. This study adhered to the principles outlined in the Declaration of Helsinki and was conducted in accordance with the guidelines of our institutional review board (No. 2022-11-046). The requirement for informed consent was waived owing to the retrospective nature of the study.

No animal subjects were used in this study.

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Abbreviations and Acronyms:
AMD = age-related macular degeneration; **ASHS-LIA** = age-related scattered hypofluorescent spots on late-phase indocyanine green angiography; **ML** = machine learning; **PCA** = principal component analysis; **PCV** = polypoidal choroidal vasculopathy; **PED** = pigment epithelial detachment; **ICGA** = indocyanine green angiography.

Keywords:
Polypoidal choroidal vasculopathy, Age-related scattered hypofluorescent spots, Indocyanine green angiography, Machine learning, Principal component analysis.

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References

1. Cheung CMG, Lai TYY, Teo K, et al. Polypoidal choroidal vasculopathy: consensus nomenclature and non-indocyanine green angiograph diagnostic criteria from the asia-Pacific Ocular imaging society PCV Workgroup. *Ophthalmology*. 2021;128:443−452.
2. Cheung CMG, Lai TYY, Ruamviboonsuk P, et al. Polypoidal choroidal vasculopathy: Definition, Pathogenesis, diagnosis, and management. *Ophthalmology*. 2018;125:708−724.
3. Choi S, Kang HM, Koh HJ. Clinical characteristics of super stable polypoidal choroidal vasculopathy after initial remission with anti-VEGF monotherapy. *Graefes Arch Clin Exp Ophthalmol*. 2021;259:837−846.
4. Kokame GT, deCarlo TE, Kaneko KN, et al. Anti-vascular endothelial growth factor resistance in exudative macular degeneration and polypoidal choroidal vasculopathy. *Ophthalmol Retina*. 2019;3:744−752.
5. Cho M, Barbazetto IA, Freund KB. Refractory neovascular age-related macular degeneration secondary to polypoidal choroidal vasculopathy. *Am J Ophthalmol*. 2009;148:70−78.e71.
6. Chen L, Zhang X, Li M, et al. Drusen and age-related scattered hypofluorescent spots on late-phase indocyanine green angiography, a candidate correlate of lipid accumulation. *Invest Ophthalmol Vis Sci*. 2018;59:5237−5245.
7. Chen L, Zhang X, Li M, et al. Age-related scattered hypofluorescent spots on late-phase indocyanine green angiography as precursor lesions of polypoidal choroidal vasculopathy. *Invest Ophthalmol Vis Sci*. 2019;60:2102−2109.
8. Lam D, Zhang X, Li H, et al. Predicting gamma passing rates for portal dosimetry-based IMRT QA using machine learning. *Med Phys*. 2019;46:4666−4675.
9. Gayathri S, Krishna AK, Gopi VP, Palanisamy P. Automated binary and multiclass classification of diabetic retinopathy using haralick and multiresolution features. *IEEE Access*. 2020;8:57497−57504.
10. Spaide RF, Jaffe GJ, Sarraf D, et al. Consensus nomenclature for reporting neovascular age-related macular degeneration data: consensus on neovascular age-related macular degeneration nomenclature study group. *Ophthalmology*. 2020;127:616−636.
11. Freund KB, Korobelnik JF, Devenyi R, et al. TREAT-AND-EXTEND regimens with anti-vegf agents in retinal diseases: a literature Review and consensus recommendations. *Retina*. 2015;35(8):1489−1506.
12. Dietterich T. Overfitting and undercomputing in machine learning. *ACM Comput Surv (Csur)*. 1995;27:326−327.
13. Maćkiewicz A, Ratajczak W. Principal components analysis (PCA). *Comput Geosci*. 1993;19:303−342.
14. Freund Y, Schapire RE. Experiments with a New Boosting Algorithm. In: *Proceedings of the 13th International Conference on Machine Learning (ICML)*. Burlington, MA: Morgan Kaufmann; 1996:148−156.
15. Silapachote P, Karuppiah DR, Hanson AR. Feature selection using AdaBoost for face expression recognition. In: Villanueva JJ, ed. *Proceedings of the Fourth IASTED International Conference on Visualization, Imaging and Image Processing - Marbella, Spain*. 2004:84−89.
16. Shen L, Bai L. *AdaBoost Gabor feature selection for classification*. *Proceedings of Image and Vision Computing*. New Zealand: Citeseer; 2004:77−83.
17. Rangini M, Jiji D. Identification of Alzheimer's disease using AdaBoost classifier. *Proceedings of the international conference on applied mathematics and theoretical computer science*. 2013:229.
18. Shakeel PM, Tolba A, Al-Makhadmeh Z, Jaber MM. Automatic detection of lung cancer from biomedical data set using discrete AdaBoost optimized ensemble learning generalized neural networks. *Neural Comput Appl*. 2020;32:777−790.
19. Breiman L. Random forests. *Mach Learn*. 2001;45:5−32.
20. Mehta P, Petersen CA, Wen JC, et al. Automated detection of glaucoma with interpretable machine learning using clinical data and multimodal retinal images. *Am J Ophthalmol*. 2021;231:154−169.
21. Yoo TK, Kim SH, Kim M, et al. DeepPDT-Net: predicting the outcome of photodynamic therapy for chronic central serous chorioretinopathy using two-stage multimodal transfer learning. *Sci Rep*. 2022;12:18689.
22. Chandra RS, Ying GS. Evaluation of multiple machine learning models for predicting number of anti-VEGF injections in the comparison of AMD treatment trials (CATT). *Transl Vis Sci Technol*. 2023;12:18.
23. Bera K, Schalper KA, Rimm DL, et al. Artificial intelligence in digital pathology - new tools for diagnosis and precision oncology. *Nat Rev Clin Oncol*. 2019;16:703−715.
24. Wulczyn E, Steiner DF, Moran M, et al. Interpretable survival prediction for colorectal cancer using deep learning. *NPJ Digit Med*. 2021;4:71.
25. Skrede OJ, De Raedt S, Kleppe A, et al. Deep learning for prediction of colorectal cancer outcome: a discovery and validation study. *Lancet*. 2020;395:350−360.
26. L'Imperio V, Wulczyn E, Plass M, et al. Pathologist validation of a machine learning-derived feature for colon cancer risk stratification. *JAMA Netw Open*. 2023;6:e2254891.
27. Chen L, Zhang X, Liu B, et al. Age-related scattered hypofluorescent spots on late-phase indocyanine green angiography: the multimodal imaging and relevant factors. *Clin Exp Ophthalmol*. 2018;46:908−915.
28. Chen L, Yang P, Curcio CA. Visualizing lipid behind the retina in aging and age-related macular degeneration, via indocyanine green angiography (ASHS-LIA). *Eye (Lond)*. 2022;36:1735−1746.
29. Tsujikawa A, Sasahara M, Otani A, et al. Pigment epithelial detachment in polypoidal choroidal vasculopathy. *Am J Ophthalmol*. 2007;143:102−111.
30. Sakurada Y, Sugiyama A, Tanabe N, et al. Choroidal thickness as a prognostic factor of photodynamic therapy with aflibercept or ranibizumab for polypoidal choroidal vasculopathy. *Retina*. 2017;37:1866−1872.
31. Jin KW, Kim JH, Park JY, et al. Long-term outcomes of ranibizumab vs. aflibercept for neovascular age-related macular degeneration and polypoidal choroidal vasculopathy. *Sci Rep*. 2021;11:14623.
32. Cho HJ, Kim KM, Kim HS, et al. Intravitreal aflibercept and ranibizumab injections for polypoidal choroidal vasculopathy. *Am J Ophthalmol*. 2016;165:1−6.